Signature Confidence Explorer

How many datasets stand behind a disease signature, and do they agree?

CMSE 830 Midterm Project | Shubham Koirala | Data: OCTAD 2.0

Who it is for, and what the data looks like

USER

A researcher about to build on a disease signature

QUESTION

Is it backed by more than one dataset, and do they agree?

DATA

OCTAD 2.0 meta analysis
350 diseases, 7.3 million rows

ONE ROW

One gene in one disease

WHAT THE DATA LOOKS LIKE (real rows)

disease	gene	log2FC	padj	I2	k
Gonorrhea	SCN2A	4.45	0.003	NA	1
Glioblastoma	ADGRG7	7.17	1.9e-11	0.0	2
Glioblastoma	RPE65	2.07	1.4e-13	0.0	16
Glioblastoma	SELENBP1	-1.00	1.3e-06	27.8	18

log2FC effect size and direction

padj significance

k datasets supporting the gene

I2 disagreement between datasets, 0 to 100 (needs 2+)

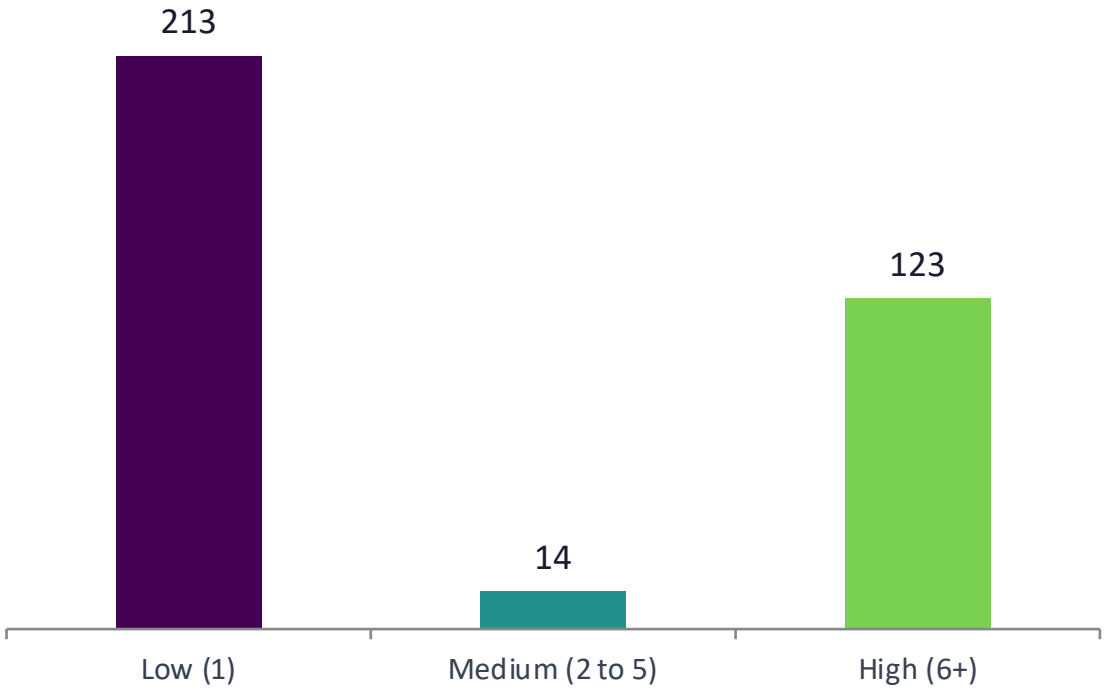
Data preparation and decisions

Check	Found	Decision
Missing padj	8.1% of rows, all in 1 dataset genes	Not imputed, gap is structural
I2 for 1 dataset	4.7M zeros that mean nothing	Set to missing
padj = 0	12,105 values	Plotted capped at 50
Skew and range	padj skewed, k up to 577	-log10 and log color scale, 3 tiers
Duplicates, extremes	Repeated symbols, |log2FC| > 10	Kept and documented
Scaling, encoding, model	Descriptive app	N/A, nothing feeds a model

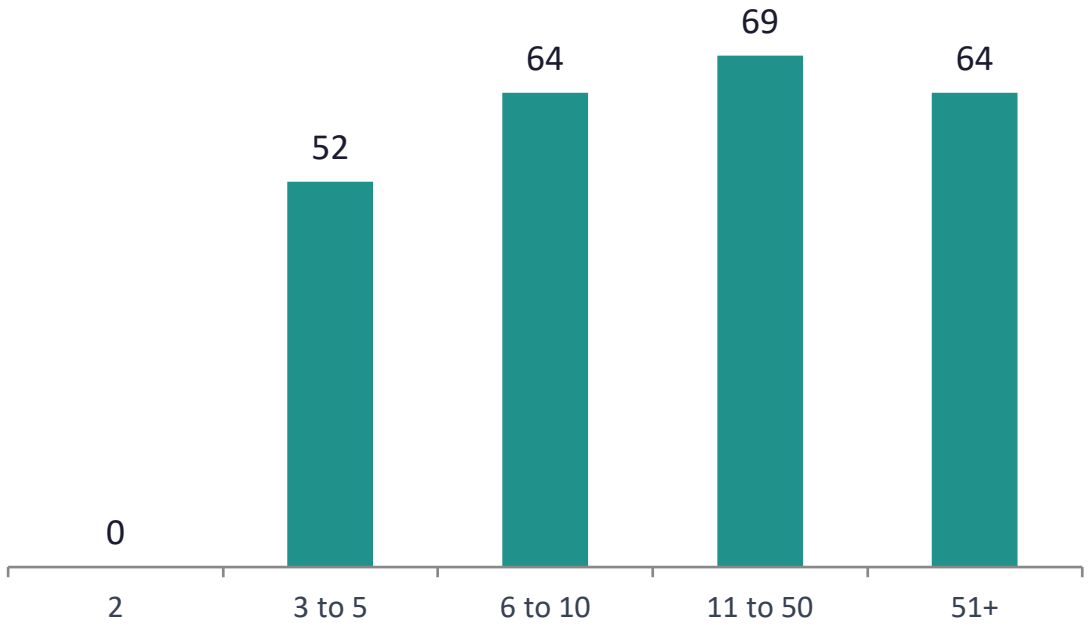


More datasets do not mean more agreement

Diseases by datasets behind their best gene



Median I^2 (disagreement, %) by datasets behind a gene



Across all genes, significance is unrelated to dataset count (rho 0.01) and effect sizes shrink (median |log2FC| 0.38 to 0.20).

The confidence card: same app, very different evidence

Gonorrhea

Confidence tier	Best supported gene	Significant genes (padj < 0.05)	Dataset disagreement (I^2)
Low	1 dataset	210	Not measurable
Low (1 dataset)	A typical gene has 1	1% of the genes that were tested	Needs 2 or more datasets

One dataset behind every gene. Nothing has replicated it, and agreement can't be measured.

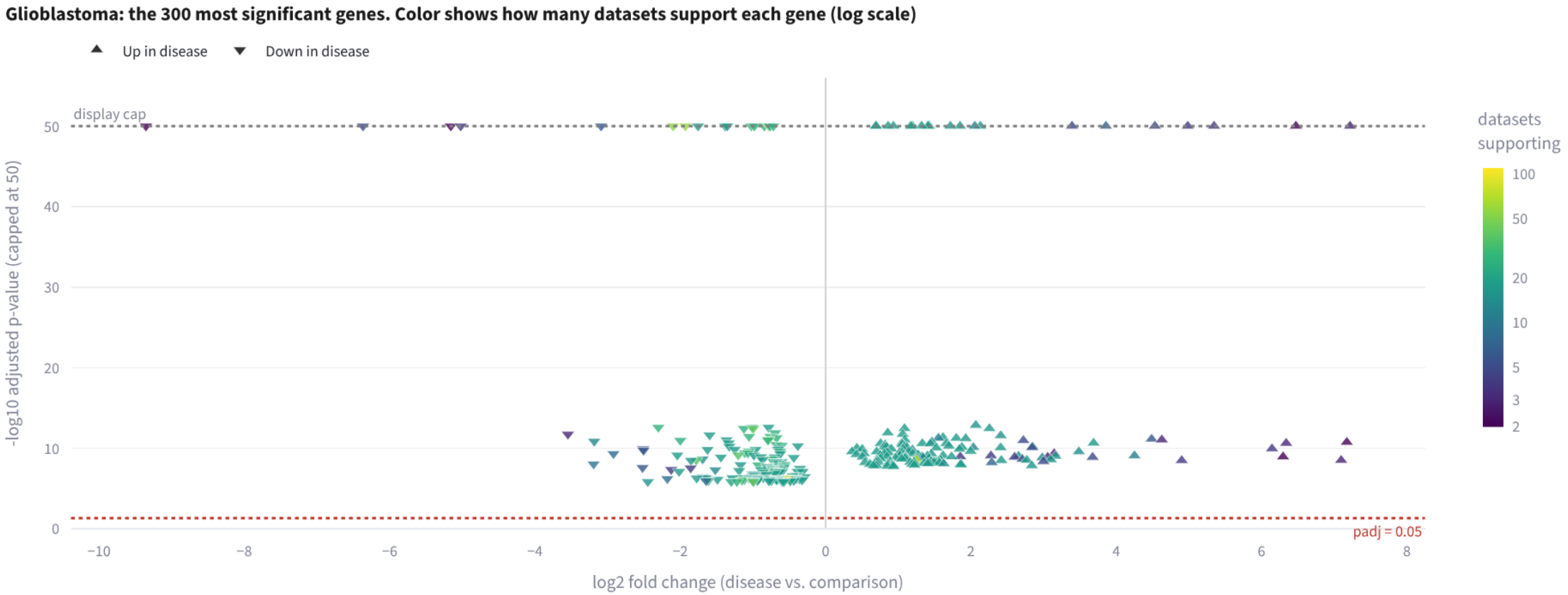
Glioblastoma

Confidence tier	Best supported gene	Significant genes (padj < 0.05)	Dataset disagreement (I^2)
High	110 datasets	7,386	Substantial
High (6 or more datasets)	A typical gene has 17	30% of the genes that were tested	Median 63%

Well supported: 66% of genes have 6+ datasets. Disagreement is substantial, so check each gene's I^2.

https://cmse830-signature-confidence.streamlit.app/

Visualization, Volcano Plot, color shows dataset support



Slider hides genes with few datasets. Genes on the top dotted line are capped and not ranked against each other.

Limits and what comes next

LIMITS

- Dataset count is a proxy for confidence, not a measurement of it.
- Only the 300 strongest genes per disease are plotted.
- Descriptive only. Tier cut points (1, 2 to 5, 6+) are my choice.

NEXT STEP

Add a pathway view of the replicated genes.

HOW TO USE IT

As a guide to check the underlying datasets, not to accept or discard a signature.

Questions and feedback

LINKS

Live app
cmse830-signature-confidence.streamlit.app

Code and README
github.com/shubham-koirala/CMSE830 (Midterm folder)